

LAST SESSION

Key Takeaways

The whole course in thirteen sentences — the things worth remembering.

One idea per slide — the things worth keeping.



An LLM only predicts the next token — your prompt is its entire working memory.

Be specific: state the role, the task, the audience and the format — specificity is control.

When words fail, show examples — a few good ones beat a page of rules.

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Common patterns — core prompting

Zero-shot

Ask directly with a precise instruction — no examples.

Few-shot

Give 2–5 input→output examples to fix the format and labels.

Role / persona

Assign an identity to set tone, vocabulary and rigor.

Delimiters

Fence the input so it reads as data, not as instructions.

Output constraints

State length, schema or allowed values up front.

No prompt change ships without a number: measure it,
don't just feel it.

Trivial wording changes can flip the output, so test a prompt's robustness on purpose.

3

Common patterns — evaluation & safety

LLM-as-judge	Grade open-ended output against a written rubric.
Prompt versioning	Track prompts and their scores like code.
Guardrails	Validate input and output; give tools least privilege.

Let the model reason step by step only where your eval set proves it actually helps.

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**Demand structured output you can validate and repair
— never scrape answers out of prose.**

4

Common patterns — reasoning

Chain-of-thought	Ask for the steps, then the answer — helps on hard tasks.
Decomposition	Break a hard task into ordered sub-questions.
Self-consistency	Sample several paths and take the majority answer.
ReAct	Interleave reasoning with tool actions and observations.
Verify / critique	Add a step that checks the answer against the constraints.

Stop polishing sentences and start engineering what fills the context window.

Ground answers in real sources and let the model say 'I don't know.'

Common patterns — context & structure

Structured output

Emit JSON to a schema, then validate and repair.

RAG / grounding

Retrieve sources and answer only from them.

Citation + refusal

Require [n] references and allow "I don't know".

Tool calling

Expose typed tools; the description drives selection.

Memory

Persist durable facts and re-inject the relevant ones.

A tool is only as good as its description — because the description is the prompt.

Common patterns — memory

Semantic memory	Durable facts about the user or world, re-injected when relevant.
Episodic memory	Records of past interactions the agent can recall.
Procedural memory	How-to routines the agent follows consistently.
Summarize / compact	Replace old turns with a running summary to save tokens.
Retrieve on demand	Index memories in a vector store; fetch only the relevant ones.
Write & forget policy	Store only durable facts; drop stale ones; never secrets or PII.

Every prompt has a bill — optimise quality per token,
not quality at any cost.

Anything in the context can be an attack, so limit what the model is allowed to do.

Models will keep changing; a clear spec, curated context and an eval set will not.

IF YOU FORGET EVERYTHING ELSE

Specify • Curate • Measure



Three habits that outlast any model.

Thank you — go build.